DATA ANALYTICS REPORT

OLA Data Analytics

End to End: SQL Analysis and Power BI Dashboard

Dataset: 103,024 ride bookings | Period: July 2024

Tools: PostgreSQL, Power BI

1. Executive Summary

This report analyses one month of OLA ride bookings (1–31 July 2024) to understand how many requests turn into completed rides, why the rest fail, and which parts of the business generate revenue. The work covers the full analytics cycle: a structured SQL table, ten business queries, an interactive Power BI dashboard, and the recommendations below.

103,024	62.1%	₹35.08M	₹21.45M
Total bookings	Success rate	Completed revenue	Value on failed bookings

Key findings

- Only 63,967 of 103,024 bookings (62.1%) became completed rides. The other 37.9% were cancelled or never matched.
- Drivers cancelled 18,434 bookings (17.9%), almost twice the 10,499 (10.2%) cancelled by customers. A further 10,124 (9.8%) found no driver.
- Booking value attached to failed bookings totals ₹21.45M, equal to about 61% of what completed rides earned (₹35.08M).
- Revenue is evenly spread across seven vehicle types (₹4.9M to ₹5.2M each). Prime Sedan is highest at ₹5.22M.
- Cash (54.8%) and UPI (40.5%) account for 95% of completed rides.
- Service quality is steady: driver and customer ratings both average 4.0 on every vehicle type.

2. Problem Statement

Ride-hailing margins depend on converting each request into a completed trip. Every cancellation wastes driver time, damages customer trust, and loses revenue. OLA has booking data with status, vehicle, location, value, payment, rating and cancellation-reason fields, but the raw table does not by itself explain where value is being lost.

Business questions

- What share of bookings succeed, and how are the failures distributed?
- Do drivers or customers cancel more, and for which reasons?
- Which vehicle types and payment methods contribute most to revenue?
- How consistent are ratings, turnaround times and ride distances?
- Which actions would raise the completion rate and revenue?

3. Solution Approach

The project follows four stages, each producing a deliverable that feeds the next.

Stage	Activity	Output
1. Collect	Load Bookings CSV (20 columns) into PostgreSQL	Table “ola” with Booking_ID as primary key
2. Query	Write ten SQL queries for counts, averages, rankings and totals	Answers to the core business questions
3. Visualise	Build Power BI dashboard (KPI cards, charts, slicers)	OLA_Analytics_Dashboard.pbix
4. Recommend	Interpret results and propose actions	This report and the presentation

Tools and technology

- PostgreSQL for table design and analysis queries.
- Power BI Desktop for the interactive dashboard, using supplied icons (cancelled, revenue, rating, vehicle, account) as KPI card graphics.
- Python (pandas, matplotlib) to verify results and draw the charts in this report.

4. Dataset Overview

Attribute	Description
Rows / columns	103,024 rows, 20 columns
Period	1 July 2024 to 31 July 2024 (31 days)
Unique customers	94,544 (most booked 5 rides; average 1.09)
Pickup locations	50 areas in Bengaluru
Vehicle types	Auto, Bike, eBike, Mini, Prime Plus, Prime Sedan, Prime SUV
Booking statuses	Success, Canceled by Driver, Canceled by Customer, Driver Not Found
Key measures	Booking_Value, Ride_Distance, V_TAT, C_TAT, Driver_Ratings, Customer_Rating

Data quality

Booking_ID has no duplicates. Rating, payment method, turnaround time and incomplete-ride fields are blank for 39,057 non-successful bookings. This is expected, because those events only occur on

completed rides, so the blanks were left as nulls and not filled. The SQL schema stores ratings as NUMERIC(2,1) and Booking_ID as the primary key.

5. Analysis and Findings

5.1 Booking outcomes

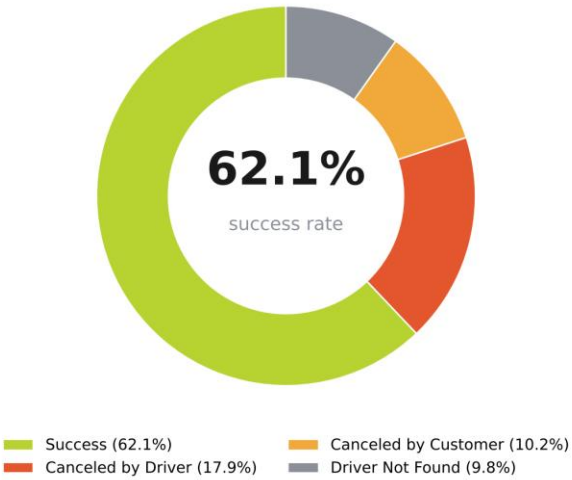


Figure 1: Booking status distribution

Status	Bookings	Share
Success	63,967	62.1%
Canceled by Driver	18,434	17.9%
Canceled by Customer	10,499	10.2%
Driver Not Found	10,124	9.8%

Roughly four in ten bookings fail. Of the 63,967 completed rides, 3,926 (6.1%) were flagged as incomplete: Customer Demand 1,601, Vehicle Breakdown 1,591, and Other Issue 734.

5.2 Revenue by vehicle type

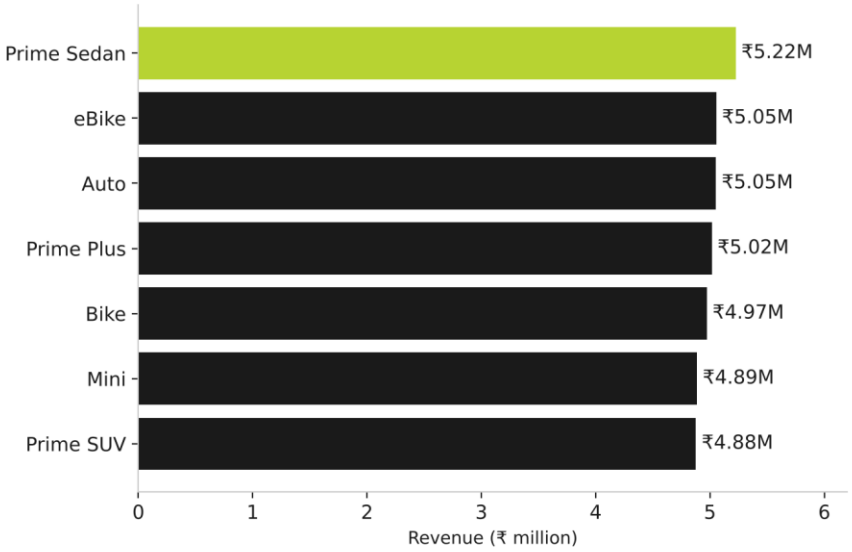


Figure 2: Completed revenue by vehicle type (₹ million)

Vehicle	Rides	Revenue (₹)	Avg value (₹)	Avg km
Prime Sedan	9,379	5,224,050	557	25.0
eBike	9,180	5,054,662	551	25.2
Auto	9,167	5,051,846	551	10.0
Prime Plus	9,075	5,015,165	553	25.0
Bike	9,134	4,971,968	544	24.9
Mini	9,036	4,885,961	541	25.0
Prime SUV	8,996	4,876,815	542	24.9

Revenue differs by only about 7% between the top and bottom vehicle types. Auto rides average 10 km against roughly 25 km for every other type, yet earn a similar value per ride (₹551), so Auto is the most distance-efficient category.

5.3 Cancellation reasons

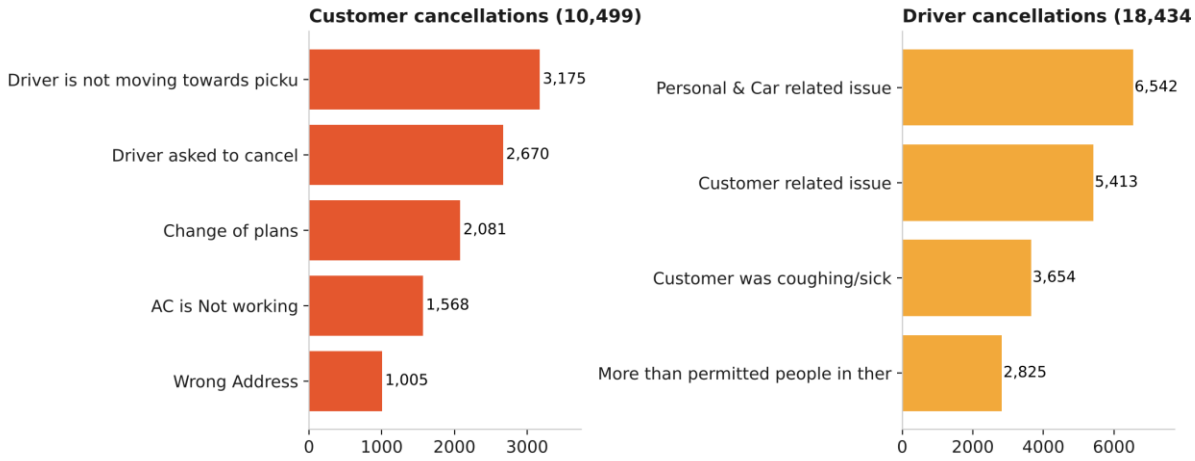


Figure 3: Cancellation reasons by customers and by drivers

Among customer cancellations, the two largest reasons relate to driver behaviour: “Driver is not moving towards pickup location” (3,175) and “Driver asked to cancel” (2,670). Together they make up 5,845 of 10,499 (56%). Among driver cancellations, “Personal & car related issue” leads with 6,542, followed by “Customer related issue” (5,413), “Customer was coughing/sick” (3,654) and “More than permitted people” (2,825).

5.4 Payment methods

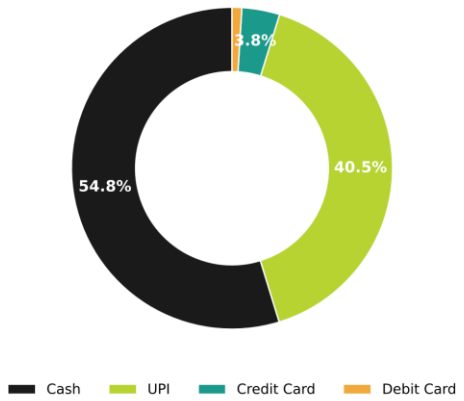


Figure 4: Payment method share of completed rides

Cash covers 35,022 rides (54.8%), UPI 25,881 (40.5%), credit card 2,435 (3.8%) and debit card 629 (1.0%).

5.5 Demand pattern

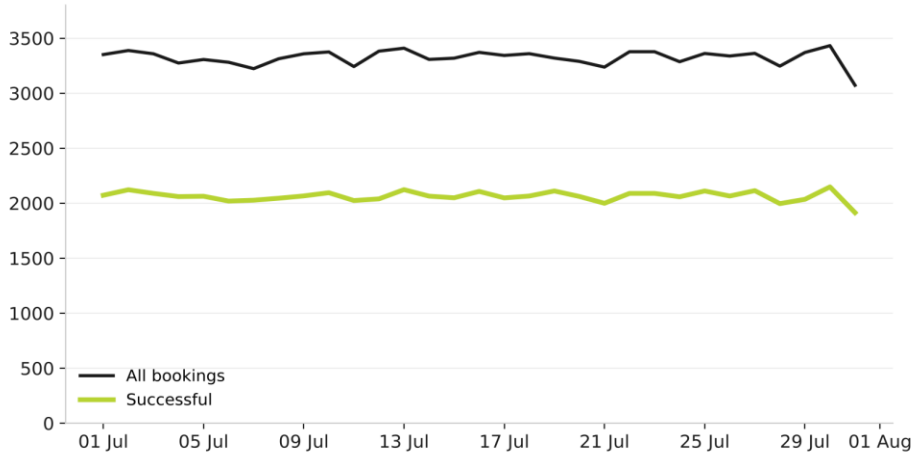


Figure 5: Daily bookings and successful rides, July 2024

Demand is remarkably steady: daily bookings range between 3,072 and 3,432 (mean about 3,323). Average bookings per weekday are similar for all seven days (3,253 to 3,385), and the failure rate holds near 38% in every hour of the day. In this dataset the leakage is therefore a structural issue, not a peak-hour supply shortage.

5.6 Service quality

Measure	Value
Average driver rating	4.00 / 5
Average customer rating	4.00 / 5
Avg vehicle turnaround (V_TAT)	170.9 sec
Avg customer turnaround (C_TAT)	84.9 sec

Ratings sit between 3.98 and 4.01 for every vehicle type, so quality is consistent. Opportunity lies in completing more rides, not in repairing satisfaction.

6. SQL Analysis Summary

Ten queries were written against the “ola” table. They are grouped below by purpose.

#	Business question	SQL technique
1	Retrieve successful bookings and count them	WHERE Booking_Status = ‘Success’
2	Average ride distance by vehicle type	AVG, GROUP BY, ORDER BY
3	Total customer cancellations	COUNT with status filter

#	Business question	SQL technique
4	Top 5 customers by number of rides	COUNT, GROUP BY, ORDER BY DESC, LIMIT 5
5	Driver cancellations: personal and car issues	GROUP BY with HAVING
6	Max and min driver rating for Prime Sedan	MAX, MIN
7	Rides paid with UPI	GROUP BY with HAVING
8	Average customer rating per vehicle type	AVG, GROUP BY
9	Total booking value of successful rides	SUM with filter
10	Incomplete rides and reasons	WHERE Incomplete_Rides = 'Yes'

7. Power BI Dashboard

The supplied Power BI file (OLA_Analytics_Dashboard.pbix) presents the findings interactively. Recommended dashboard layout: headline KPI cards (total bookings, success rate, revenue, average distance, ratings), a booking status breakdown, revenue and rides by vehicle type, cancellation reason charts for customers and drivers, payment mix, and slicers for date, vehicle type and pickup location.

8. Recommendations

Area	Action	Evidence
Driver reliability	Monitor drivers who cancel or do not head to pickup; add warnings and incentives for acceptance and completion	18,434 driver cancellations; 5,845 customer cancellations tied to driver behaviour
Supply coverage	Offer incentives in pickup zones with frequent "Driver Not Found"	10,124 unmatched requests
Fleet health	Vehicle checks for AC and breakdowns; clearer in-app policy on passenger limits	1,568 AC cancellations; 1,591 breakdowns; 2,825 over-capacity cancellations
Address quality	Add pin-drop confirmation to cut wrong-address cancellations	1,005 wrong address cancellations
Digital payments	Reward UPI and card use for cash riders	Cash is 54.8% of rides

Impact estimate: **recovering 10% of the ₹21.45M lost on failed bookings would add about ₹2.1M a month.**

9. Conclusion

The OLA booking data shows a business with stable demand, balanced revenue across vehicle types and healthy ratings, but with 37.9% of bookings failing to complete. Driver-side cancellations and unmatched requests are the clearest levers. Targeted driver quality controls, supply incentives and fleet checks give the largest and most measurable gain for the least change.

Next steps

- Add location-level cancellation analysis to find problem zones.
- Collect timestamps for cancellations to measure response delays.
- Re-run this analysis monthly to track improvement.